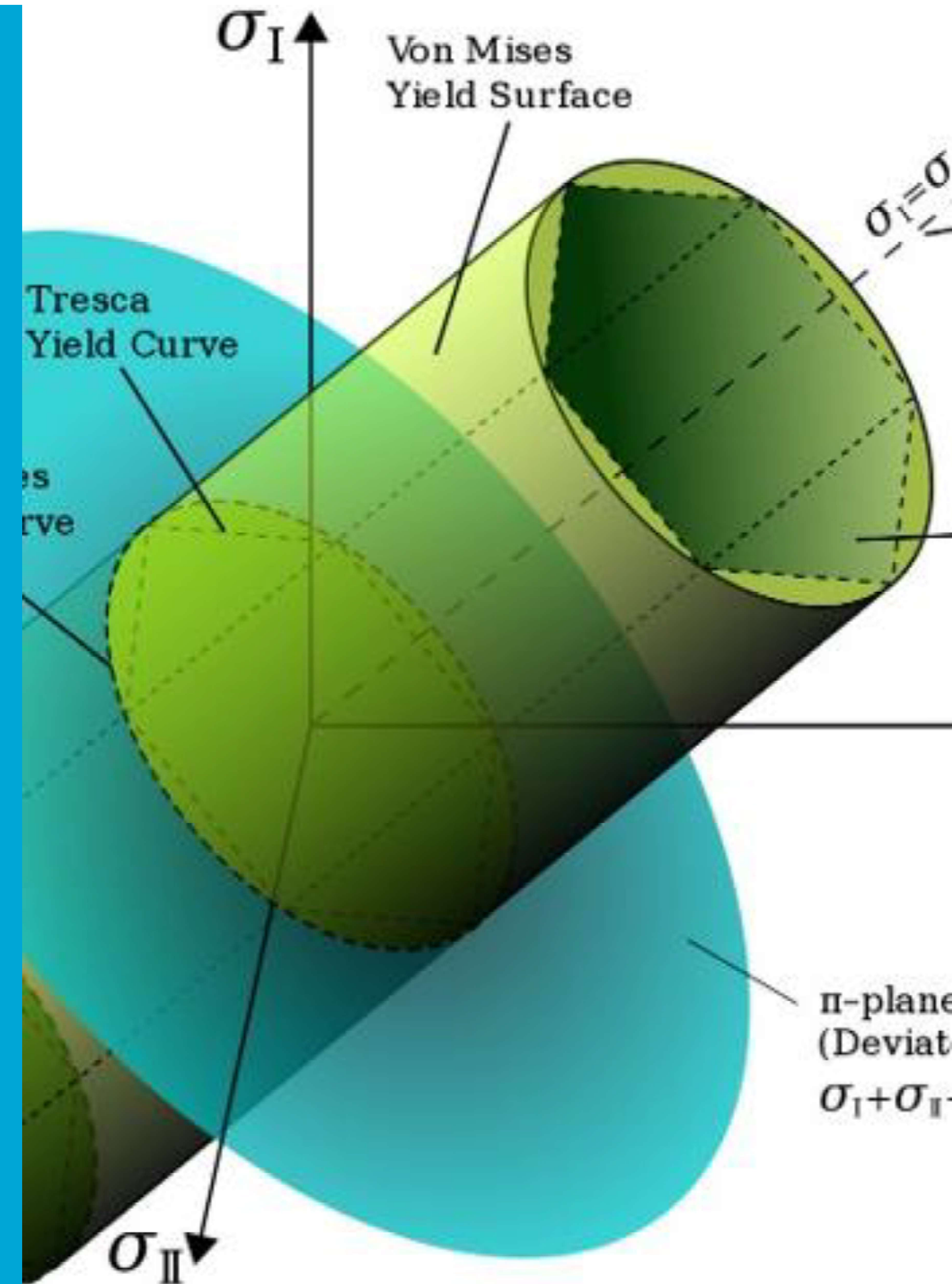


Theory of elasticity

1, 2, and 7 September 2026

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Learning objectives

At the end of the lectures, the student is able to:

1. Define the **strain and stress tensor** at a point
2. Apply **transformation rules** to strain and stress tensors
3. Compute **principal strains and stresses**
4. Relate strains and stresses for a **linear elastic** material
5. Assess the **failure of linear elastic materials** according to the most common failure criteria for both ductile and quasi-brittle materials

Scope of the lectures

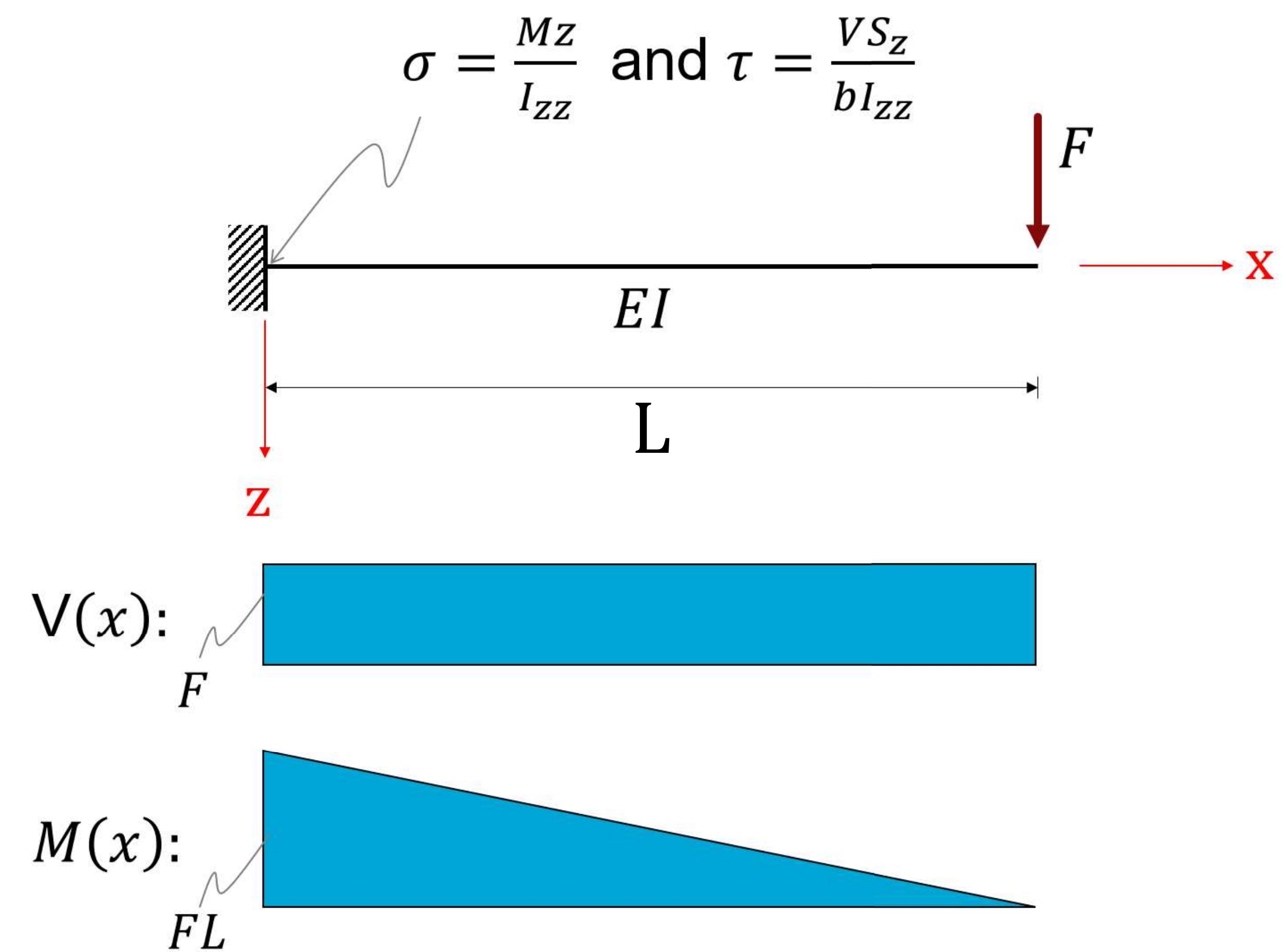
- How can we calculate what is the safety coefficient before failure?
- How can we design a safe structure?

The lectures of the week provide **the basis** to answer these two questions by explaining how to compute **failure at material level**.

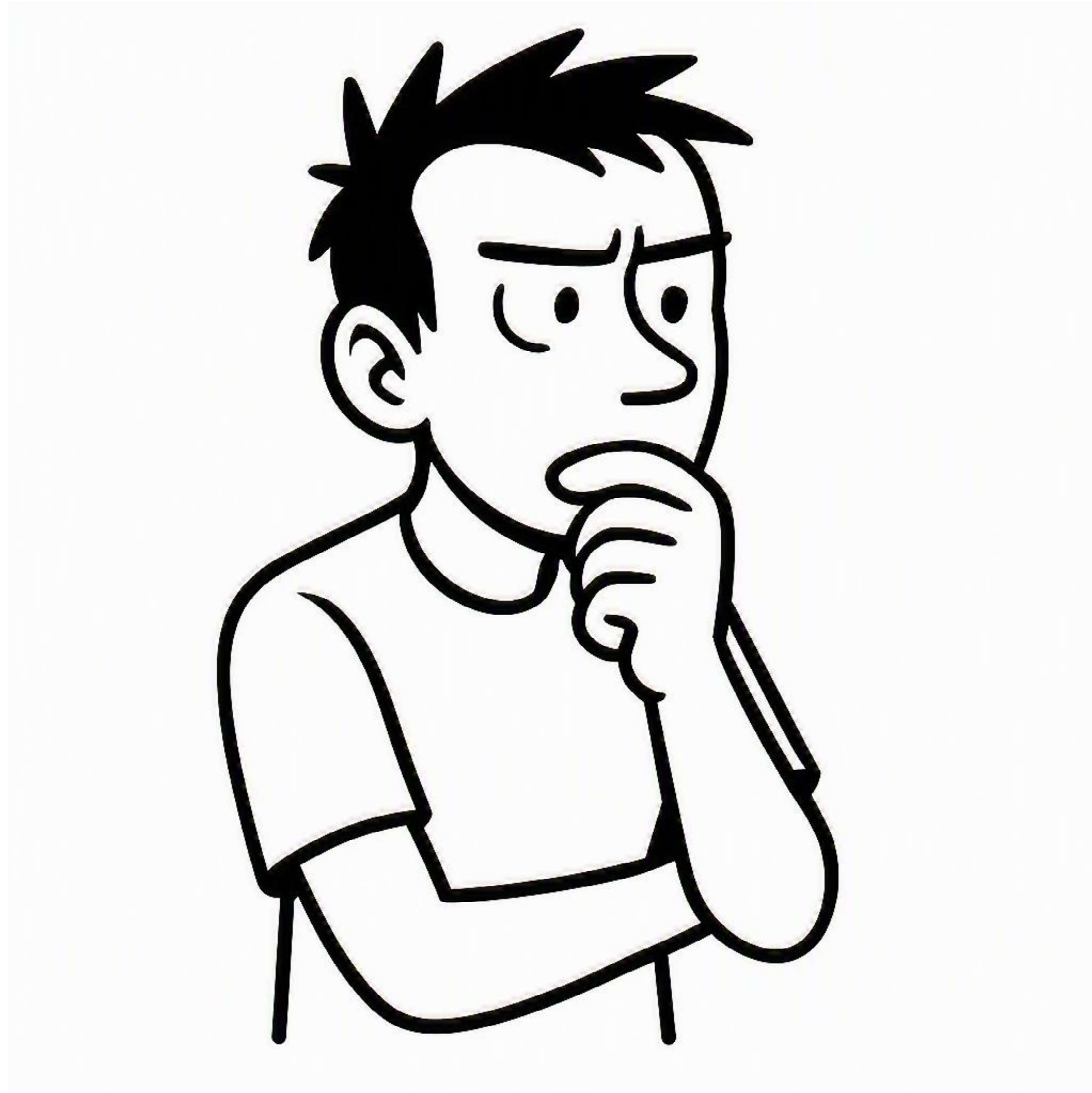
01

Stresses and strains

What do we know already?

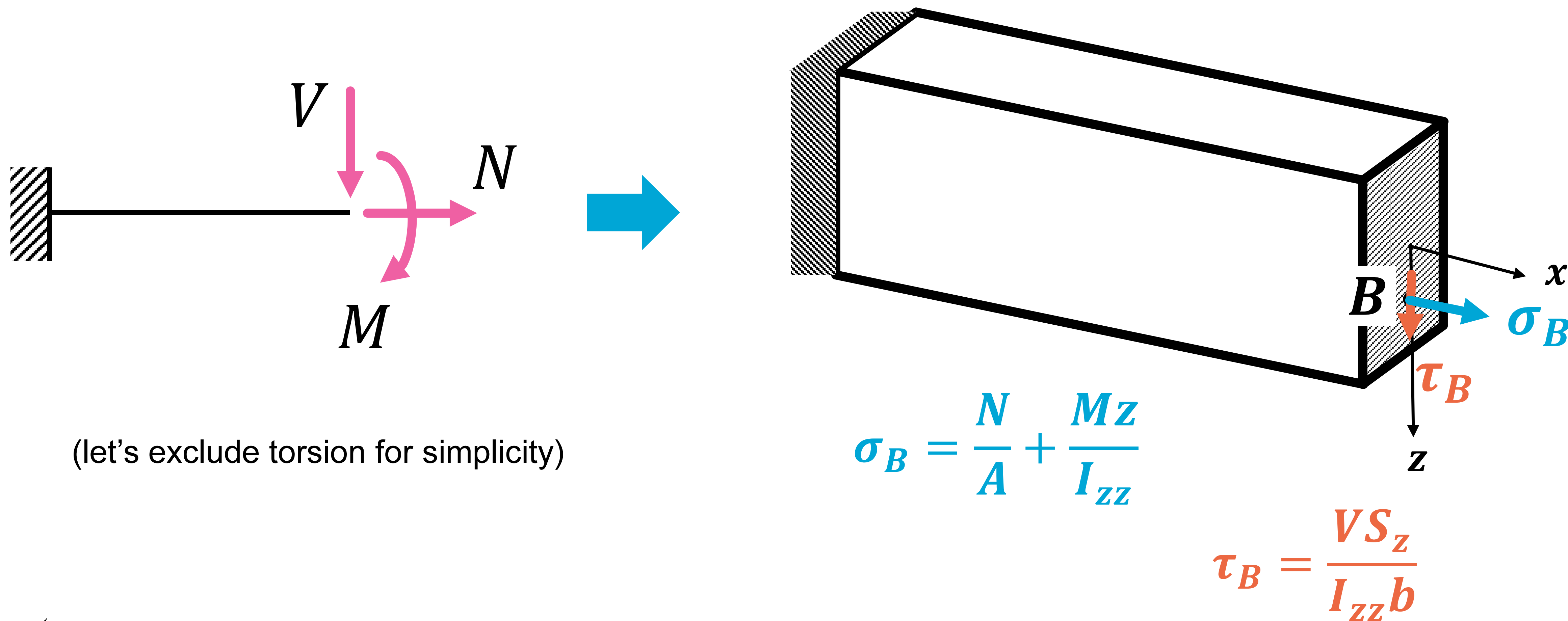


What we don't know yet!

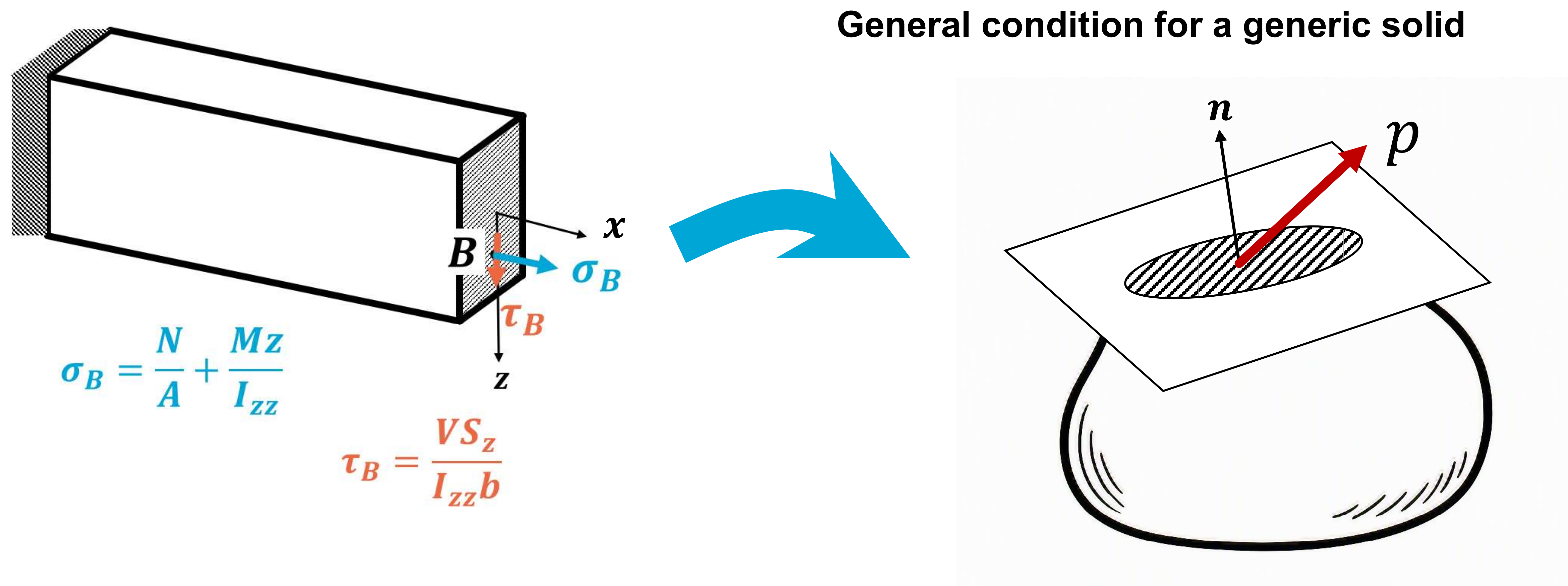


From stresses in beam theory to 3D stress-state

Stresses in a beam for the Euler-Bernoulli theory are only a special condition



From stresses in beam theory to 3D stress-state



- What is the stress state of point B in a generic solid when an external stress is applied under an angle to a surface cutting the solid?
- How can we express this stress state?

Some notation...

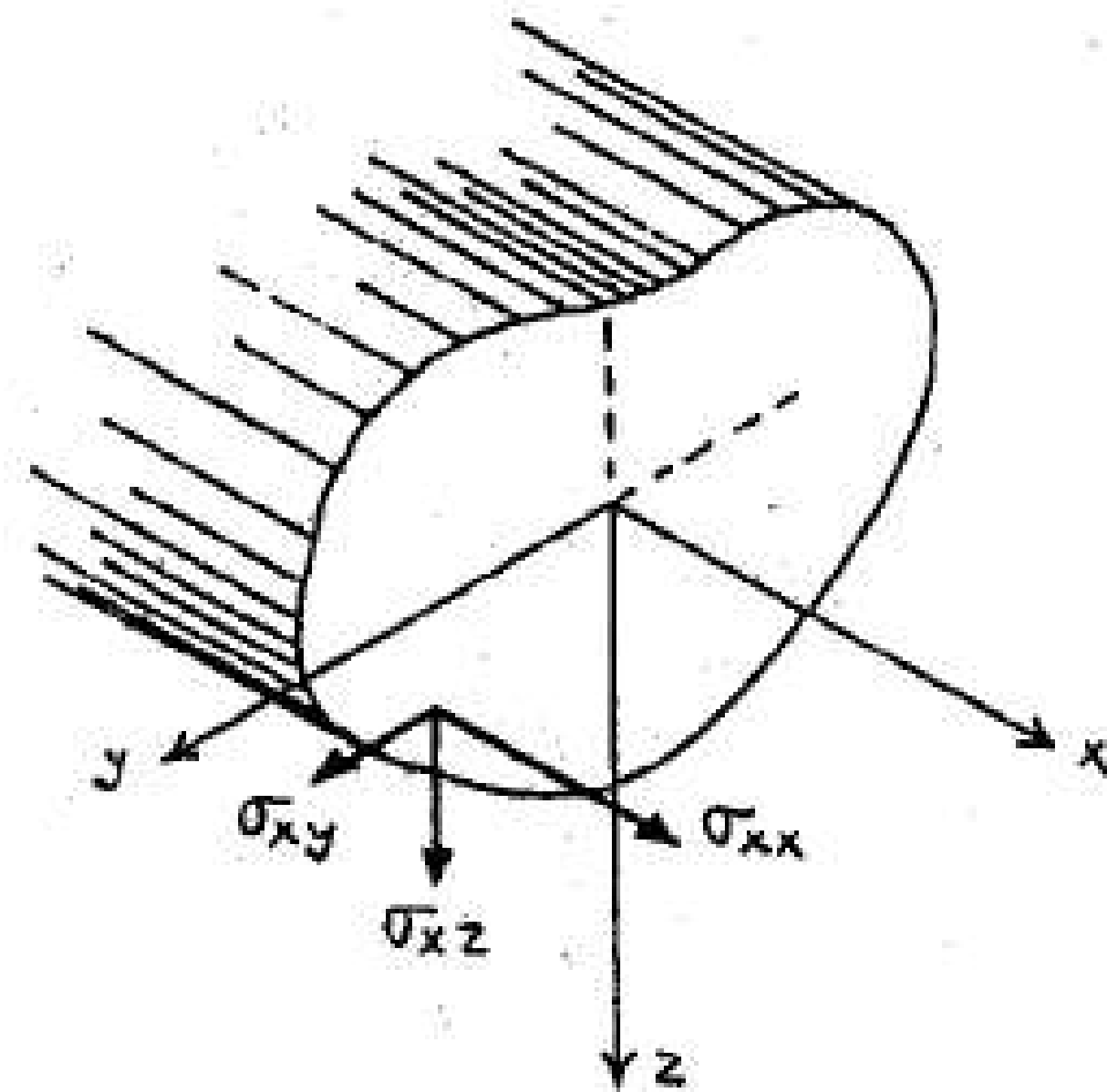
A plane is identified via the vector acting the normal direction (e.g. the x-plane coincides with the plane whose axes in the plane are y and z – see image below)

The stress is then noted via a double index:

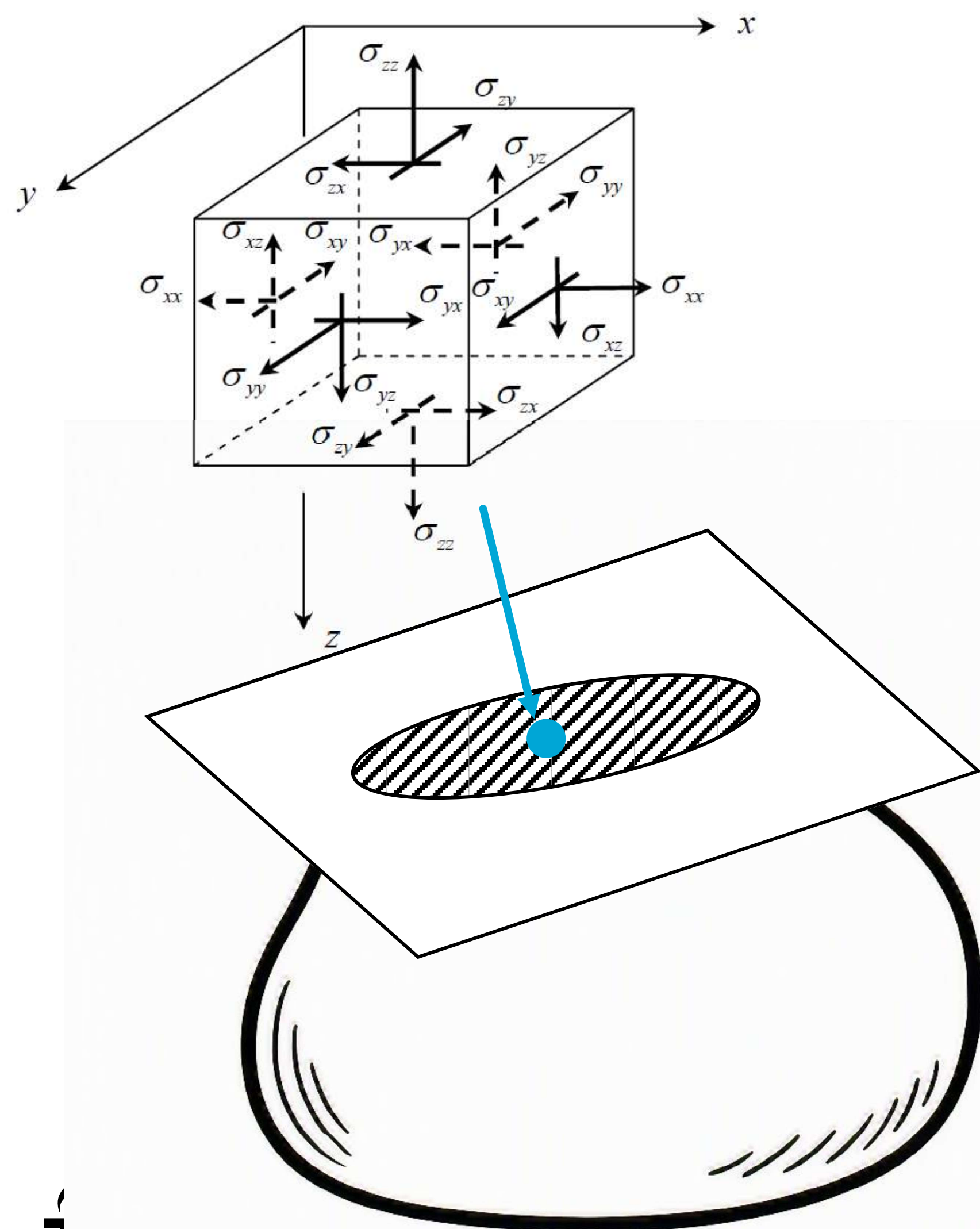
E.g. σ_{xy}

x = stress in the x-plane

y = along the y-direction



From stresses in beam theory to 3D stress-state



To characterize the **stress state in a point**, we need to know all stress components!

From 2 (σ , τ) to **6 stresses**:

- **3 normal stresses** (along the axes x, y, z): σ_{xx} , σ_{yy} , σ_{zz}
- **3 shear stresses** (acting on the planes perpendicular to axes x, y, z): σ_{xy} , σ_{xz} , σ_{yz}

These stresses are organized in the **stress tensor**:

$$\begin{bmatrix} \sigma_{xx} & \sigma_{xy} & \sigma_{xz} \\ \sigma_{xy} & \sigma_{yy} & \sigma_{yz} \\ \sigma_{xz} & \sigma_{yz} & \sigma_{zz} \end{bmatrix}$$

Note: **the tensor is symmetric!**

3D stress tensor

We want to answer 3 questions:

1. What is a tensor?
2. What is the stress tensor?
3. Why do we need a tensor to express the stress state at a point?

3D stress tensor

We want to answer 3 questions:

1. What is a tensor?

A tensor is a mathematical object that generalizes scalars, vectors, and matrices to higher dimensions. Examples:

- 0th-order tensor: **scalar** (e.g. temperature)
- 1st-order tensor: **vector** (e.g. force, displacement)
- 2nd-order tensor: **matrix** (e.g. stress tensor)

2. What is the stress tensor?

3. Why do we need a tensor to express the stress state at a point?

3D stress tensor

We want to answer 3 questions:

1. What is a tensor?
2. What is the stress tensor?

The stress tensor is a **3 × 3 matrix** (for a **3D problem**) that describes the **complete stress state at a point** in a material. It contains:

- **Normal stresses** (acting perpendicular to a plane): $\sigma_{xx}, \sigma_{yy}, \sigma_{zz}$
- **Shear stresses** (acting parallel to a plane): $\sigma_{xy}, \sigma_{xz}, \sigma_{yz}$

Although there are 9 components, only **6 components are independent** because the **tensor is symmetric** ($\sigma_{ij} = \sigma_{ji}$ with $i, j = x, y, z$)

3. Why do we need a tensor to express the stress state at a point?

3D stress tensor

We want to answer 3 questions:

1. What is a tensor?
2. What is the stress tensor?
3. Why do we need a tensor to express the stress state at a point?

The **stress at a point** cannot be described by a single value because it **depends on the orientation of the plane passing through that point**. For each plane orientation, the stress vector has a different magnitude and direction, with different normal and shear components. Therefore, to fully describe the stress state at a point, **we need a quantity that relates the orientation of a plane to the stress acting on that plane**. This quantity is **the stress tensor**.

Isotropic and deviatoric stresses

- The stress tensor can be split into a diagonal tensor with identical terms and a non-diagonal tensor:

$$\sigma = \begin{bmatrix} \sigma_{xx} & \sigma_{xy} & \sigma_{xz} \\ \sigma_{xy} & \sigma_{yy} & \sigma_{yz} \\ \sigma_{xz} & \sigma_{yz} & \sigma_{zz} \end{bmatrix} = \underbrace{\begin{bmatrix} \sigma_0 & 0 & 0 \\ 0 & \sigma_0 & 0 \\ 0 & 0 & \sigma_0 \end{bmatrix}}_{\text{isotropic stress}} + \underbrace{\begin{bmatrix} \sigma_{xx} - \sigma_0 & \sigma_{xy} & \sigma_{xz} \\ \sigma_{xy} & \sigma_{yy} - \sigma_0 & \sigma_{yz} \\ \sigma_{xz} & \sigma_{yz} & \sigma_{zz} - \sigma_0 \end{bmatrix}}_{\text{deviatoric stresses}}$$

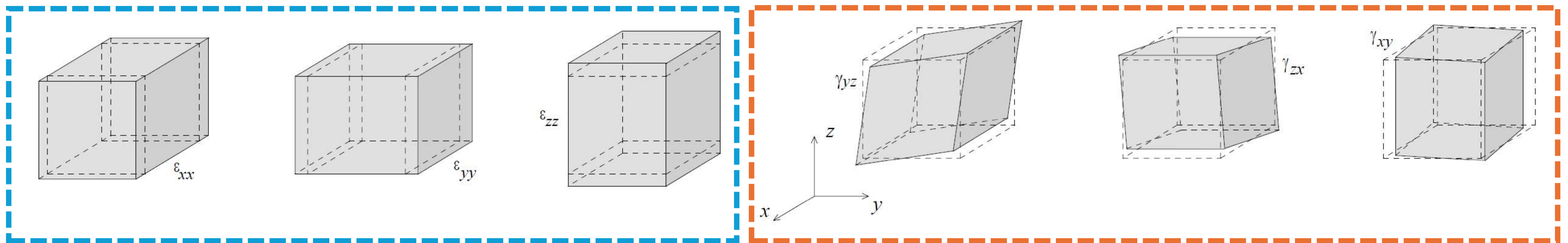
- The **isotropic stress** $\sigma_0 = \frac{1}{3}(\sigma_{xx} + \sigma_{yy} + \sigma_{zz})$ causes a **change in volume** only.
- The **deviatoric stress** components cause a **distorsion**.

We need this distinction when we introduce the failure criteria for ductile materials (metals).

3D Strain tensor

A structure under external loads deforms. Both a **change in volume** and **in shape** is possible. The **amount of deformation per unit length** is called specific elongation or **strain**.

As for the stresses, in 3D we have **6 independent strains**: ϵ_{xx} , ϵ_{yy} , ϵ_{zz} , γ_{xy} , γ_{zx} , γ_{yz}



How can we calculate the strains from observable displacements?

Strains and displacements: general 3D case

As for the stresses, the deformation at a material point can be expressed via the **strain tensor**:

$$\begin{bmatrix} \epsilon_{xx} & \epsilon_{xy} & \epsilon_{xz} \\ \epsilon_{xy} & \epsilon_{yy} & \epsilon_{yz} \\ \epsilon_{xz} & \epsilon_{yz} & \epsilon_{zz} \end{bmatrix}$$

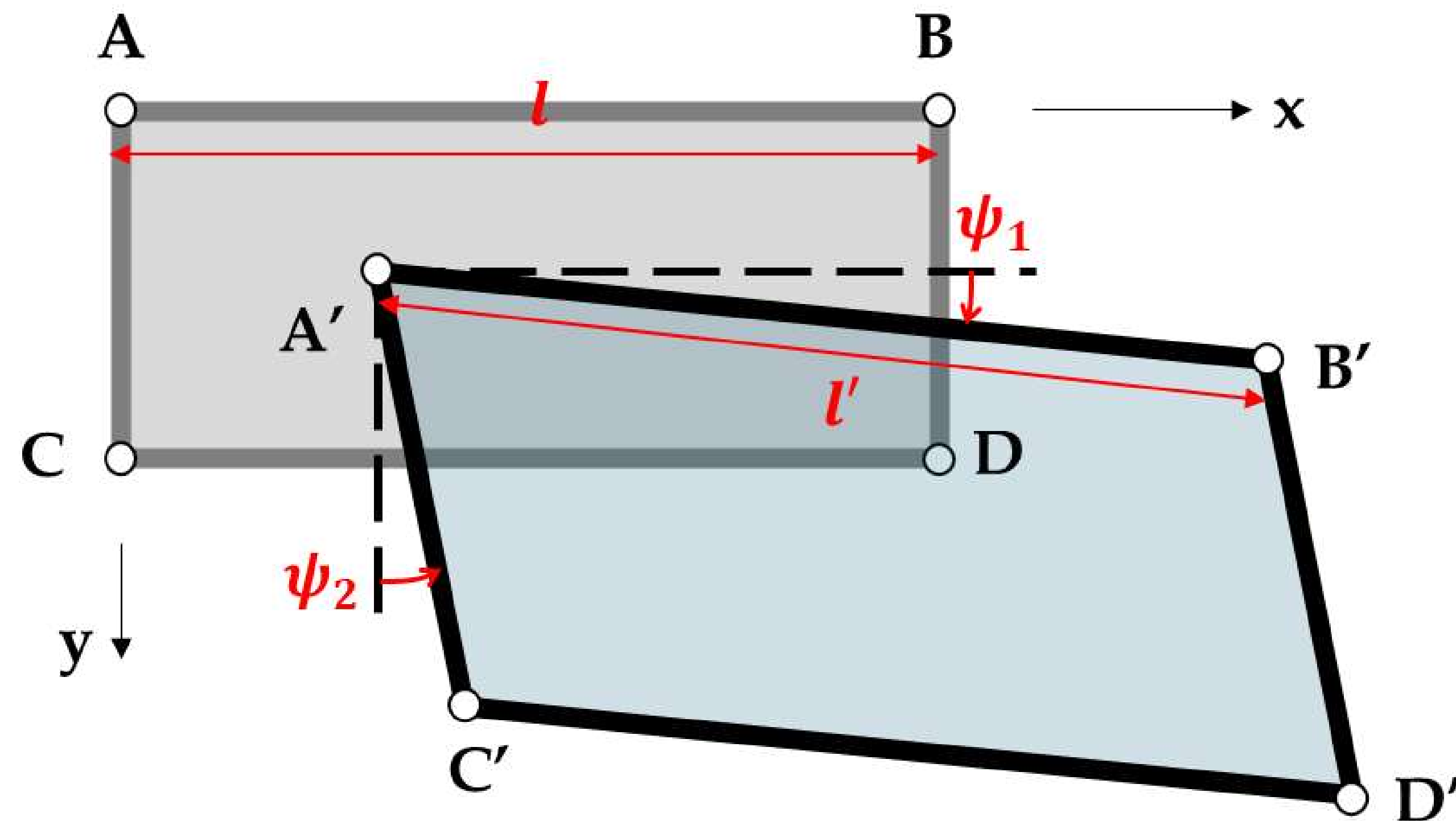
Also in this case, **the tensor is symmetric** and **there are 6 independent components**.

How to compute the strain components from displacements?

$$\epsilon_{ij} = \frac{1}{2} \left(\frac{\partial u_i}{\partial j} + \frac{\partial u_j}{\partial i} \right), \text{ with } i, j = x, y, z$$

E.g., in the 2D case: $\epsilon_{xx} = \frac{\partial u_x}{\partial x}$; $\epsilon_{yy} = \frac{\partial u_y}{\partial y}$; $\epsilon_{xy} = \epsilon_{yx} = \frac{1}{2} \left(\frac{\partial u_x}{\partial y} + \frac{\partial u_y}{\partial x} \right)$

Strains and displacements: shear deformation



We define the **shear deformation γ** as the total change of the right angle between the x- and y-fibres (2D case):

$$\gamma = \psi_1 + \psi_2$$

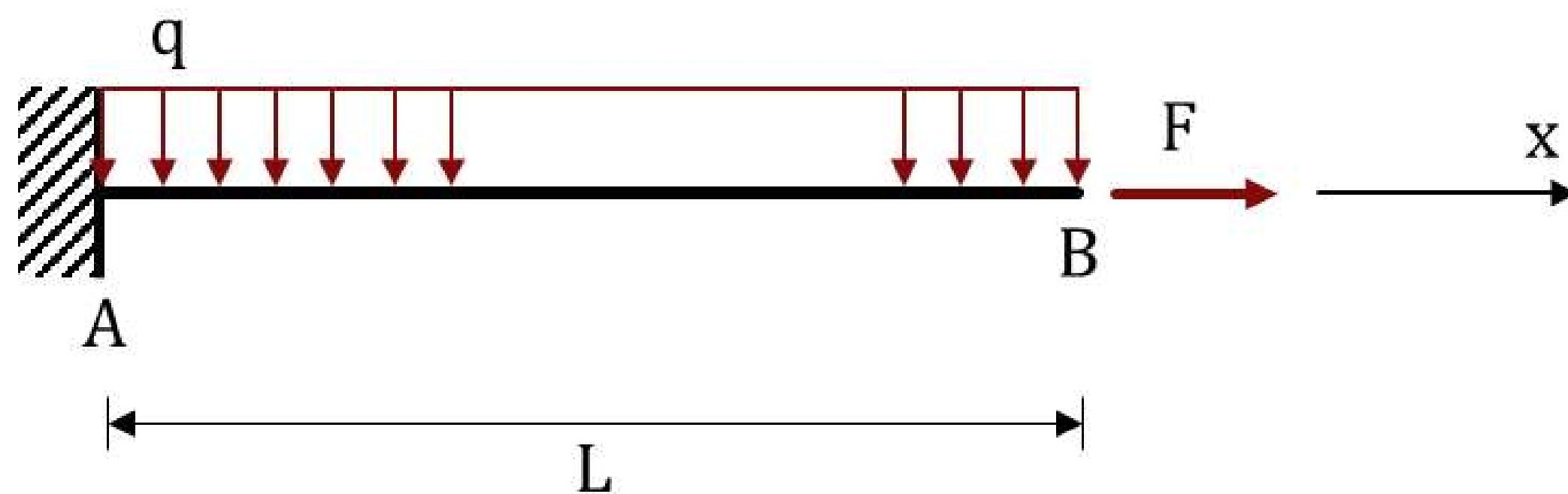
The shear deformation is equal to twice the off-diagonal terms of the strain tensor:

$$\gamma_{xy} = 2\varepsilon_{xy}$$

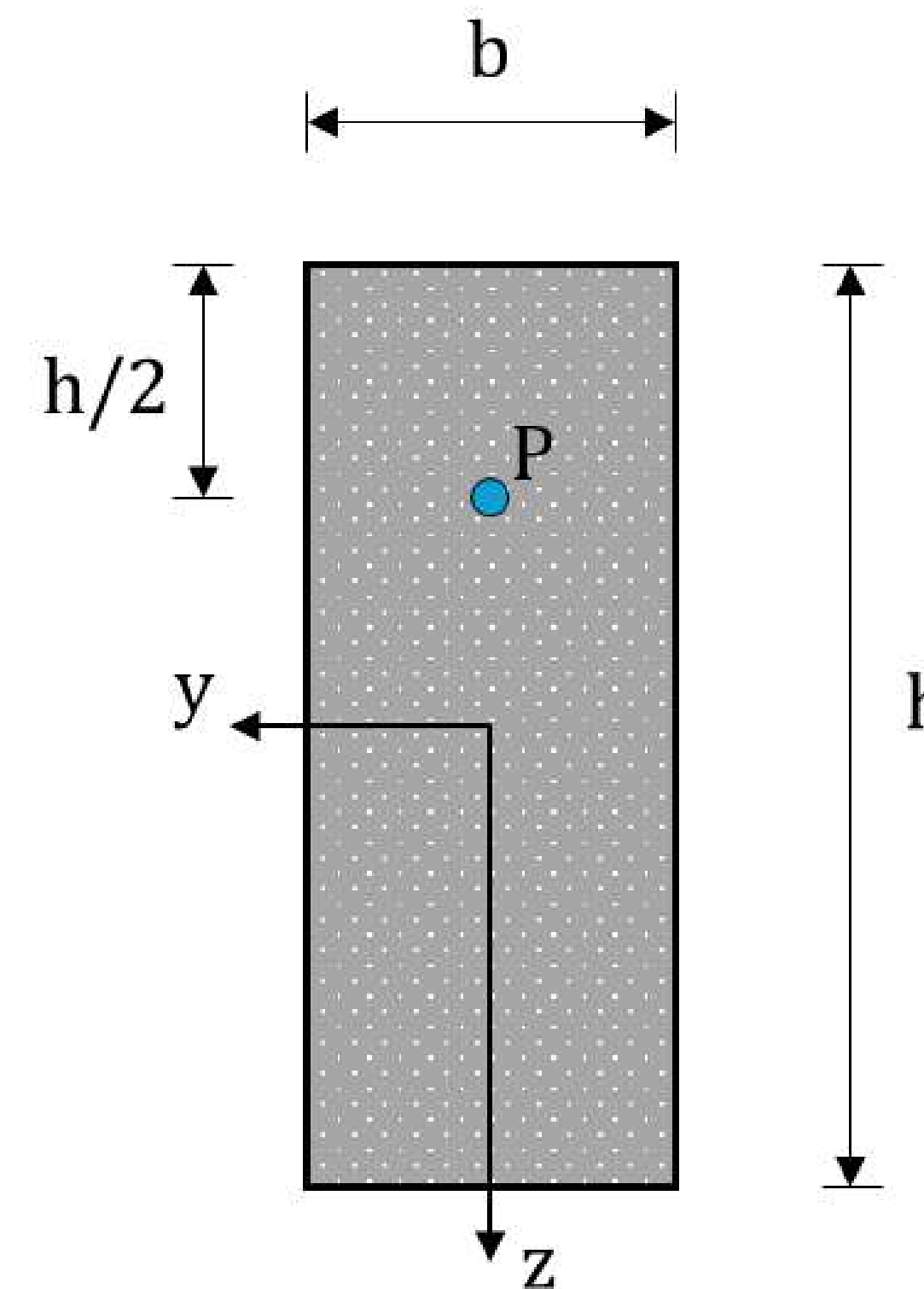
$$\gamma_{xz} = 2\varepsilon_{xz}$$

$$\gamma_{yz} = 2\varepsilon_{yz}$$

Example #01



Rectangular section:



$$\begin{aligned} L &= 4 \text{ m} \\ E &= 30000 \text{ MPa} \\ \nu &= 0.2 \\ q &= 20 \text{ kN/m} \\ F &= 200 \text{ kN} \end{aligned}$$

$$\begin{aligned} b &= 200 \text{ mm} \\ h &= 600 \text{ mm} \end{aligned}$$

For the given beam, compute the stress matrix (tensor) in the xyz-coordinates at the point P of section A-A, and the isotropic and deviatoric components.

Example #01 – solution procedure

- Internal forces at section A-A:

$$N_A = F = 200 \text{ kN}$$

$$V_A = ql = 80 \text{ kN}$$

$$M = -\frac{ql^2}{2} = -160 \text{ kNm}$$

Example #01 - solution procedure

- Stresses:

- $\sigma_{yy} = \sigma_{xy} = \sigma_{yz} = \sigma_{zz} = 0$

- The normal stress along the beam axis is equal to:

$$\sigma_{xx}(P) = \frac{Mz(P)}{I_{zz}} + \frac{F}{A} = \frac{M(-h/4)}{I_{zz}} + \frac{F}{A} = 8.33 \text{ MPa}$$

Where: $A = 1.2 \cdot 10^5 \text{ mm}^2$ and $I_{zz} = 3.6 \cdot 10^9 \text{ mm}^4$

- The shear stress:

$$\sigma_{xz}(P) = \frac{V_A S_z^*(P)}{I_{zz} b} = 0.75 \text{ MPa}$$

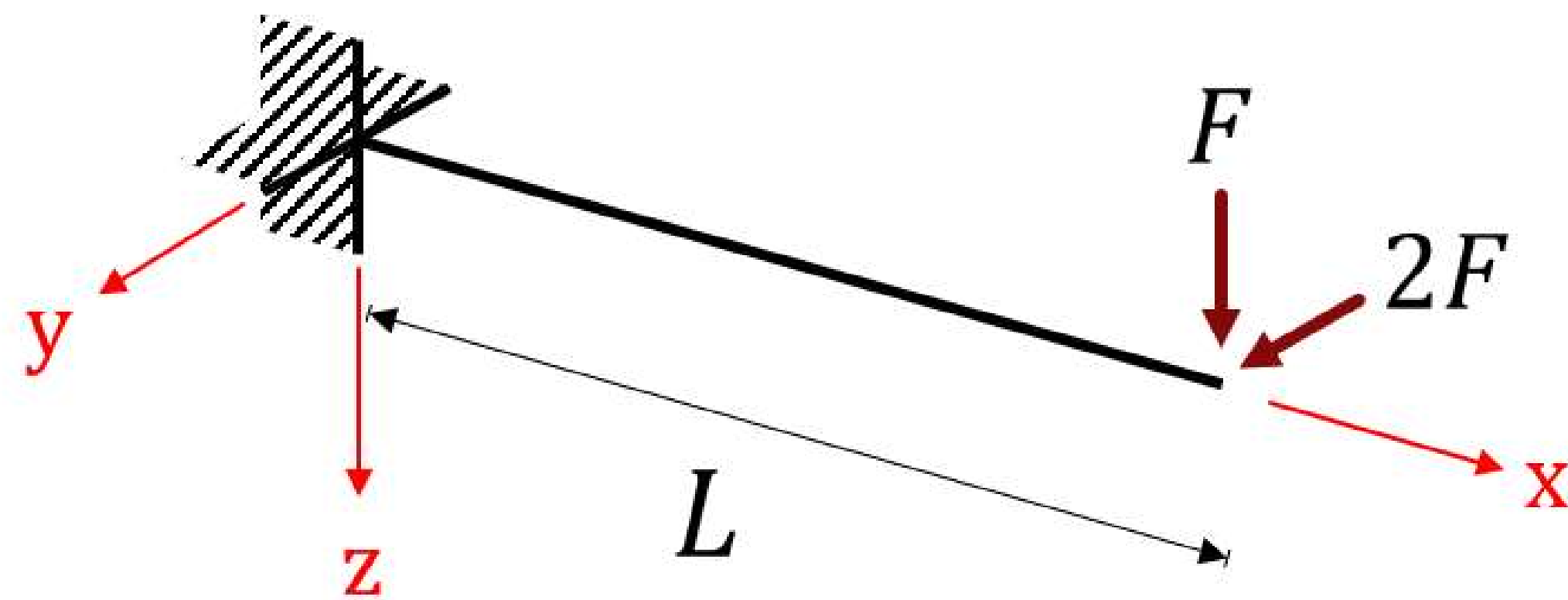
Where: $S_z^*(A) = \frac{1}{4} h b \cdot \frac{3}{8} h = \frac{3}{32} h^2 b = 6.75 \cdot 10^6 \text{ mm}^3$

Example #01 - solution procedure

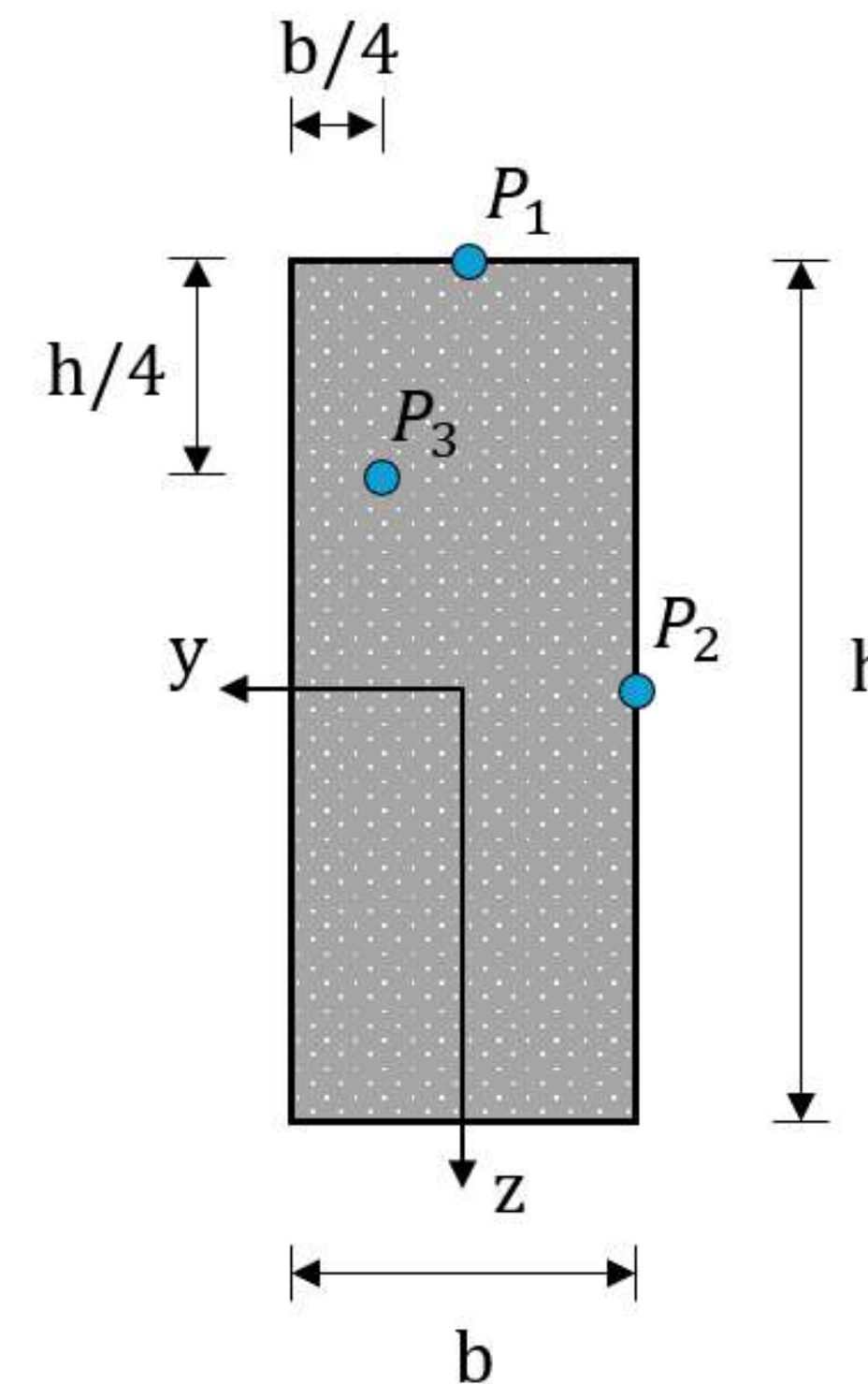
- The isotropic stress is: $\sigma_0 = \frac{\sigma_{xx}}{3} = 2.77 \text{ MPa}$
- The stress tensor in xyz-coordinates is:

$$\sigma = \begin{bmatrix} 8.33 & 0 & 0.75 \\ 0 & 0 & 0 \\ 0.75 & 0 & 0 \end{bmatrix} \text{ MPa} = \underbrace{\begin{bmatrix} 2.77 & 0 & 0 \\ 0 & 2.77 & 0 \\ 0 & 0 & 2.77 \end{bmatrix}}_{\text{Isotropic components}} \text{ MPa} + \underbrace{\begin{bmatrix} 5.55 & 0 & 0.75 \\ 0 & -2.77 & 0 \\ 0.75 & 0 & -2.77 \end{bmatrix}}_{\text{Deviatoric components}} \text{ MPa}$$

Guided exercise #01



Rectangular section:



$$\begin{aligned} L &= 4 \text{ m} \\ E &= 30000 \text{ MPa} \\ \nu &= 0.2 \end{aligned}$$

$$F = 5 \text{ kN}$$

$$\begin{aligned} \text{Rectangular} \\ \text{section:} \\ b &= 200 \text{ mm} \\ h &= 600 \text{ mm} \end{aligned}$$

For the given beam, compute the stress matrix (tensor) in the xyz-coordinates at the point P_1 , P_2 , and P_3 of section A-A, and the isotropic and deviatoric components.